

MAXIMAL

ARCHITECTURAL FREEDOM (SPECIFICATION STRIPPED AWAY)

MINIMAL

(c) AI-Native (Layer 3)

THE MOONSHOT HORIZON: ZERO STRUCTURAL TEMPLATE

System Intent

Workload graph
PPA limits & golden spec

Cross-Layer Closed-Loop Exploration

Compiler / Schedule

Custom Vector ISA

Carry-Save ALU

Streaming FIFOs

Physical Oracles

STA • DRC • LEC • CDC

Signed-Off Silicon

Structural container stripped away ↑

(b) AI-Driven (Layer 2)

PARAMETER SEARCH IN IMMUTABLE CONTAINER (Θ)

Fixed Workload

Target binary & constraints

Fixed Architectural Template Θ Cache θ TAGE θ 

Macro (x,y)

Closed-Loop Search

Simulator / EDA sweep

Optimal Config θ^*

Parameters automated ↑

(a) AI-Assisted (Layer 1)

POINT ASSISTANCE IN FROZEN CONTEXT

Human Task Query

"Draft branch predictor"
within frozen SoC

Human-Frozen Architecture

Fetch

AI Leaf

Decode

Execute

Mem

WB

Isolated Candidate

Drafted code snippet

Manual Integration